

Data & Analytics Layer BRD

1. Summary

A centralized, cloud-based reporting dashboard designed to provide Vibe House management with actionable operational and revenue intelligence . The module aggregates data from the eZee PMS, Razorpay payment gateway, and the internal Staff Task Dashboard to monitor financial performance, track staff efficiency, and prevent internal leakage and shrinkage.

2. Use Case: Manager Analytics & Inventory Control

2.1 Overview

Managers and supervisors access this dashboard via a secure web portal (desktop or tablet). While mostly used for reading real-time reports, it also contains active workflows for managing physical assets (marking items as damaged).

2.2 Detailed Action Sequence (Manager Workflow)

Step	User Action (Click/Type)	System Response / Backend Logic
1. Auth	Log into Vibe House Admin Portal with Manager credentials.	Validate role-based access control (RBAC); Grant session access to the Analytics module.
2. Navigation	Click "Revenue & Ops Analytics" tab.	Aggregate and fetch live data from eZee PMS API and internal Staff Task DB .
3. Data Filtering	Select Date Range (e.g., "Last 7 Days") and Department.	Query DB; Refresh graphs for Daily Add-on Revenue, SLA compliance %, and Peak Request timings.
4. Staff Review	Click on "Staff Performance Scoring".	Calculate and display total SLAs successfully completed by each staff member within the selected timeframe.

5. Inventory View	Click "Inventory Control" tab.	Fetch real-time unit-level stock counts (Opening, Sold, Damaged, Remaining) for Locks, Towels, etc .
6. Mark Damaged	Click "Report Damaged" next to a specific Unit ID (e.g., Towel-045).	Update unit status to DAMAGED in DB. Instantly decrement "Remaining" stock count by 1. Log timestamp & reporting user ID.
7. Low-Stock Alert	<i>(No User Action - Background Process)</i>	System calculates Remaining vs Threshold . If stock drops below the threshold, display a persistent RED banner on the UI.
8. Alert Dispatch	<i>(No User Action - Background Process)</i>	Trigger automated Email Alert detailing the depleted item and current count to the pre-configured Procurement Distribution List (DL).

3. Revenue Analytics Module (Financial Intelligence)

Objective: Track and visualize detailed Gross Revenue generated outside of standard room bookings.

Key Reports & Metrics:

- **Daily Add-on Revenue:** A day-by-day aggregate of all gross revenue generated from commodities (towels, locks) and services (laundry).
- **Item-wise Sales Performance:** A breakdown showing which specific items are driving the most volume and revenue.
- **Conversion Rate of Upsells:** The percentage of guests who successfully complete an add-on purchase.
- **Late Checkout Revenue Contribution:** Specific tracking of gross revenue generated purely from pre-booked and same-day late checkout requests.
- **Upgrade Revenue Tracking:** Revenue generated from Dorm-to-Private room upgrades and lower bunk upgrades.

4. Operational Analytics Module (Efficiency Tracking)

Objective: Measure how effectively the property is being run and identify bottlenecks in service delivery.

Key Reports & Metrics:

- Average Response Time per Department: Tracks the mean time taken from task creation (Pending) to completion for Housekeeping, Maintenance, and Front Office.
- SLA Compliance %: The percentage of total tasks completed within their allotted SLA timer without requiring escalation.
- Staff Performance Scoring: A quantitative score assigned to each staff member, calculated strictly based on the total number of SLAs successfully completed within the reporting period.
- Peak Request Timing Analysis: Heatmaps showing what times of day generate the highest volume of service requests, allowing for optimized staff shift scheduling.
- Dorm vs Private Service Patterns: A comparative analysis of service requests generated by dorm guests versus private room guests to better forecast inventory and staffing needs.

5. Inventory Control Module (Asset Management)

Objective: Provide granular tracking of physical assets to completely prevent internal leakage and shrinkage. Tracked Categories: Locks, Towels, Toiletry kits, and Extra mattresses .

5.1 Unit-Level Traceability

Instead of just bulk counting, the system will support Unit-Level IDs (e.g., Towel-001, Lock-045) for durable/reusable items.

- Data Points: Each unit will have an associated ID, exact pricing/cost, current status (Available, Sold, In-Use, Damaged), and a traceability log (which guest/room last used it).

5.2 Inventory Metrics Dashboard The dashboard will display the following metrics per category :

- Opening Stock: The count at the beginning of the day/shift.
- Sold / Rented: Units checked out to guests (synced with successful Razorpay/PMS transactions).
- Damaged: Units flagged as unusable.
- Remaining: Real-time available stock.